

Ivan Zhang

Clemson, SC | (917)-868-5273 | sanchonq@gmail.com

EDUCATION

Clemson University, Computer Science, B.S.

Expected Graduation: Fall 2024 - Fall 2027

PROJECTS

FEVER Counterfactual Pretraining Experiment (Python)

- Prepared 54,000 balanced FEVER records for controlled experiments by resolving evidence and checking normalized claims for overlap across training, validation, and test splits.
- Built a ~6 GB CC-News corpus toward a 900M estimated whitespace-token target across 2016–2018; applied exact-hash/MinHash exclusions and retained provenance manifests to support reproducible data preparation.
- Orchestrated dependent Slurm jobs and froze a 32,000-token BPE vocabulary plus two single-token verdict markers to standardize preprocessing and verdict representation.

Multimodal Media Forensics (Python)

- Developed a pipeline using Qwen3 Omni-Thinking and Qwen2.5 72B-Instruct to assess tone and intent in social media videos by combining visual, audio, and transcript analysis.
- Converted unstructured video into structured JSON with transcripts, audio features, and speaker labels, enabling a two-part analysis of direct video input and extracted evidence.

EXPERIENCE

Undergraduate Research — Watt AI Media Forensics

Spring 2026–Current

- Communicated multimodal research methods and findings through a poster explaining how video, audio, and transcript evidence support assessments of tone and intent.
- Examined how decoding settings, token budgets, and conversation history affect LLM responses through Python/Jupyter experiments with Clemson RCD models.
- Investigated tool-calling agents, MCP, and multimodal evaluation to identify application safeguards and distinguish valid model outputs from evidence-supported claims.

Capstone — Azure Email Knowledge Pipeline

Current

- Designed a three-layer Azure architecture to make email knowledge searchable and traceable: unchanged originals in Bronze, parsed JSON in Silver, and validated corpus data in Gold.
- Mapped automated mailbox intake and queued processing with Azure Logic Apps, Blob Storage, Event Grid, Storage Queue, and Functions to reduce reliance on manual email uploads.
- Specified a custom extraction schema and validation approach for Azure OpenAI and Claude, with PostgreSQL/pgvector as the proposed corpus store, to link extracted facts to source evidence.
- Planned permission-aware retrieval using Microsoft Entra ID and PostgreSQL row-level security to limit answers to authorized evidence; coordinated design work in a four-member Agile team through a shared backlog

SKILLS

Languages & data: Python, C/C++, SQL, data pipelines, JSON, Jupyter

Tools & platforms: AWS, Azure, Git/GitHub, UNIX, Slurm, REST APIs, OpenAPI Specification, JUnit, IntelliJ, VS Code

Additional: Agile collaboration; Mandarin (working proficiency)